COMPUTER NETWORKS LAB

SEMESTER PROJECT

“BSE (4-B)”

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<u>Title:</u> <u>Smart Traffic Management System</u>

➢ Introduction:

This project simulates a smart traffic management system using Cisco Packet Tracer. It is divided into three key areas:

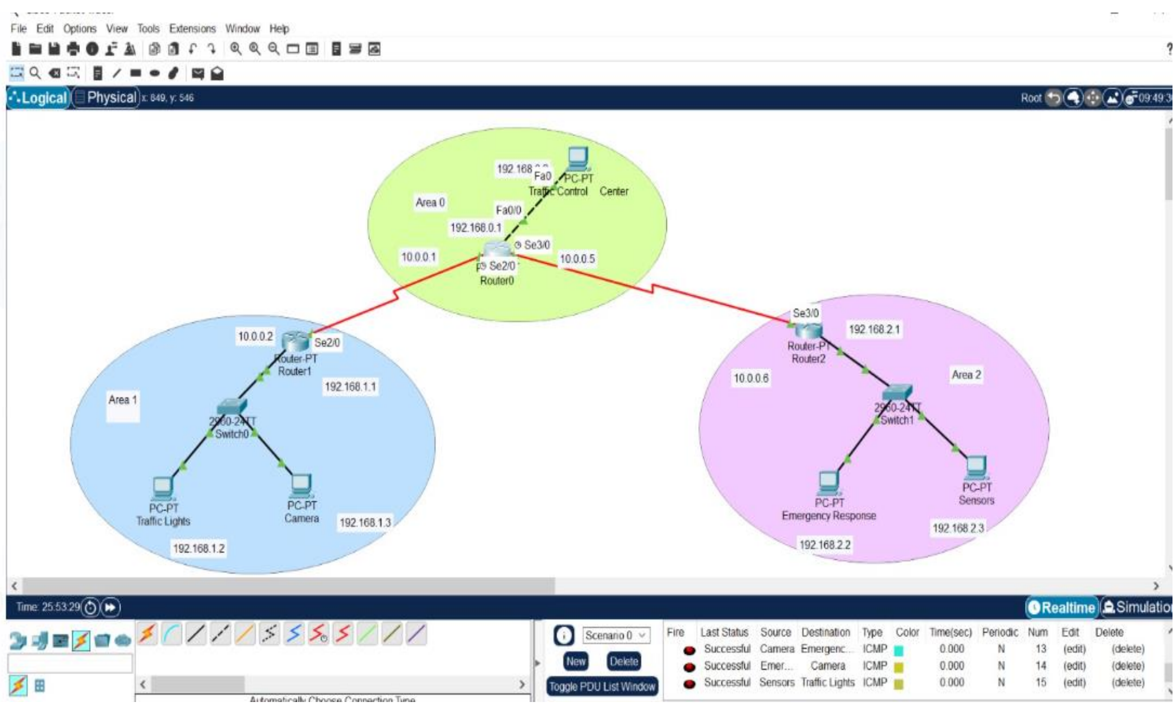
- Area 0 (Backbone Area): Traffic Control Center
- Area 1 (City Intersection A): Traffic Lights and Surveillance Cameras
- Area 2 (City Intersection B): Emergency Units and Sensors

Objective:
To create a scalable, secure, and efficient network using routers, switches, and end devices connected via OSPF routing protocol.

➢ Network Topology Design

Network Devices Used:

- 3 Routers (Router0, Router1, Router2)
- 3 Switches (Switch0, Switch1, Switch2)
- End Devices: PCs, Traffic Lights, Cameras, Sensor

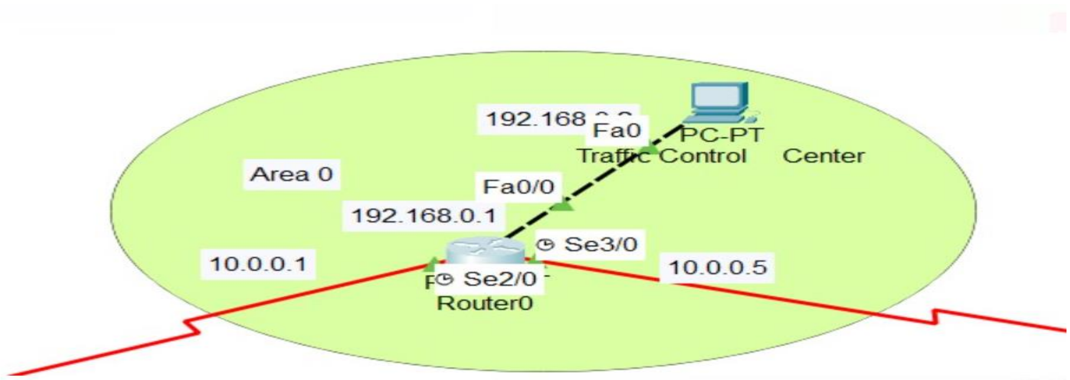


➢ Network Design Overview

The entire topology is divided into **three main areas**:

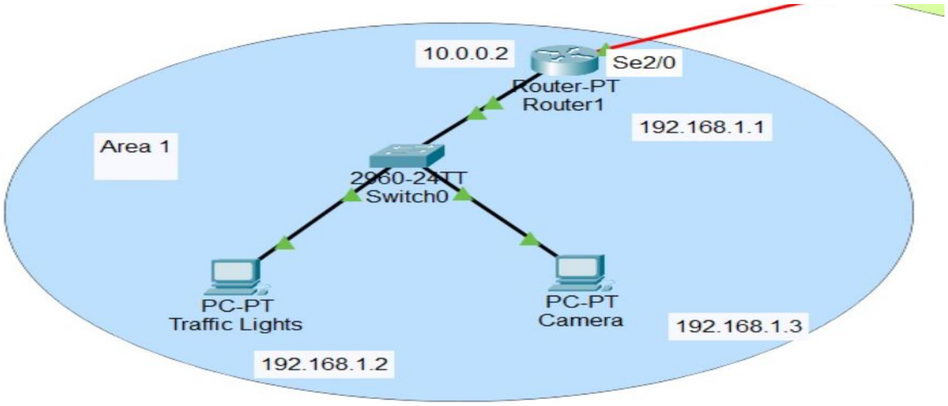
Area 0: Backbone (Core Network)

- **Device(s):** Router0 (Core router), PC0 (Traffic Control Center)
- **IP Addressing:**
    - Router0 (Interface to Area 1): 192.168.0.1
    - Router0 (Interface to Area 2): 10.0.0.5
    - PC0 (Traffic Control Center): 192.168.0.2
- **Function:** Acts as the central backbone that interconnects Area 1 and Area 2. It routes data between remote segments and the Traffic Control Center.



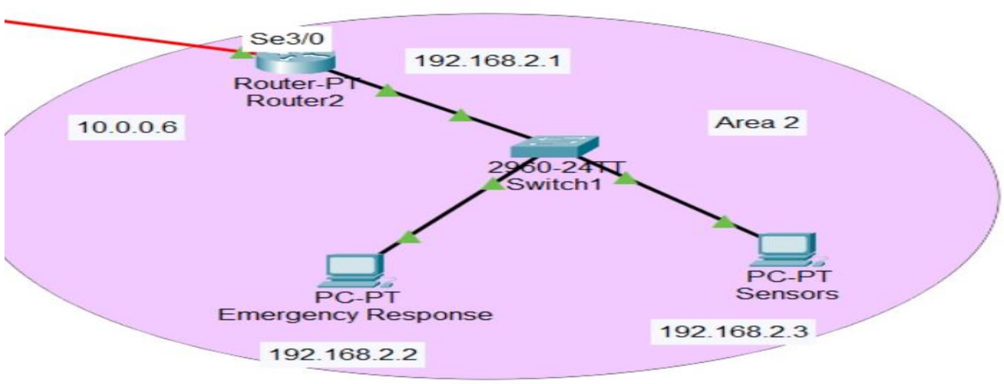
Area 1: Traffic Monitoring Zone

- **Device(s):** Router1, Switch1, PC1 (Traffic Signal), PC2 (Camera)
- **IP Addressing:**
    - Router1 (Interface to Area 0): 10.0.0.2
    - Router1 (LAN Interface): 192.168.1.1
    - PC1 (Traffic Signal): 192.168.1.2
    - PC2 (Camera): 192.168.1.3
- **Function:** Manages local traffic devices such as smart signals and surveillance cameras. Data from these devices are sent to the Traffic Control Center via Router1 and Router0.



Area 2: Emergency Response Zone

- **Device(s):** Router2, Switch2, PC3 (Emergency Response Node), PC4 (Sensor)
- **IP Addressing:**
  - Router2 (Interface to Area 0): 10.0.0.6
  - Router2 (LAN Interface): 192.168.2.1
  - PC3 (Emergency Response Node): 192.168.2.2
  - PC4 (Sensor): 192.168.2.3
- **Function:** Supports emergency scenarios. The Emergency Node can trigger alerts and communicate with the Traffic Control Center in real time. The sensor could represent air quality, temperature, or accident detection.



➢ Routing protocol used

Smart Traffic Management System, the routing protocol you have used is OSPF (Open Shortest Path First)

Why we used OSPF:

1. **Dynamic Routing:**

OSPF is a **dynamic routing protocol**, which means routers automatically learn and share routes. You don't have to manually update the routing table like in **static routing**.

2. Supports Multi-Area Design:

Your network is divided into:

- **Area 0** (Backbone – Traffic Control Center)
- **Area 1** (Intersection A – Traffic Lights & Cameras)
- **Area 2** (Intersection B – Emergency Services)

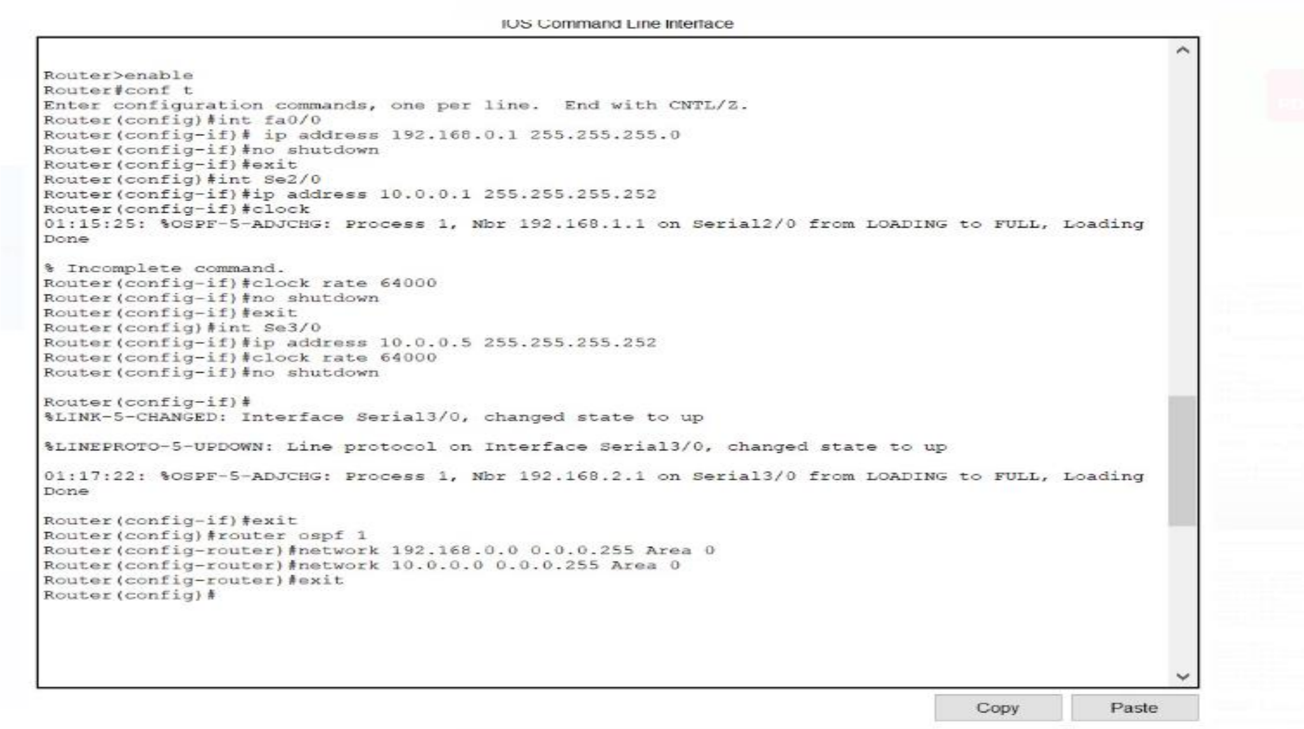
OSPF is perfect for this **multi-area architecture**. It keeps routing efficient and organized.

3. Faster Convergence:

OSPF quickly detects network changes and recalculates the best routes — important in emergency systems or smart infrastructure where **quick data routing** is needed.

➢ <u>Router Configuration:</u>

<u>Router0:</u>


IOS Command Line Interface

```
Router>enable
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#int fa0/0
Router(config-if)# ip address 192.168.0.1 255.255.255.0
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#int Se2/0
Router(config-if)#ip address 10.0.0.1 255.255.255.252
Router(config-if)#clock
01:15:25: %OSPF-5-ADJCHG: Process 1, Nbr 192.168.1.1 on Serial2/0 from LOADING to FULL, Loading
Done

% Incomplete command.
Router(config-if)#clock rate 64000
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#int Se3/0
Router(config-if)#ip address 10.0.0.5 255.255.255.252
Router(config-if)#clock rate 64000
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface Serial3/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial3/0, changed state to up

01:17:22: %OSPF-5-ADJCHG: Process 1, Nbr 192.168.2.1 on Serial3/0 from LOADING to FULL, Loading
Done

Router(config-if)#exit
Router(config)#router ospf 1
Router(config-router)#network 192.168.0.0 0.0.0.255 Area 0
Router(config-router)#network 10.0.0.0 0.0.0.255 Area 0
Router(config-router)#exit
Router(config)#
```

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<u>Router1:</u>

```
Router>enable
Router#conf  t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#int fa0/0
Router(config-if)#ip address 192.168.1.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

Router(config-if)#exit
Router(config)#int Se2/0
Router(config-if)#ip address 10.0.0.2 255.255.255.252
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface Serial2/0, changed state to up
exit
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial2/0, changed state to up

Router(config)#router ospf 1
Router(config-router)#network 192.168.1.0 0.0.0.255 Area 1
Router(config-router)#network 10.0.0.0 0.0.0.255 Area 0
Router(config-router)#exit
Router(config)#
```

Router2:

```
Would you like to enter the initial configuration dialog? [yes/no]: no


Press RETURN to get started!


Router>enable
Router# conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#int fa0/0
Router(config-if)#ip address 192.168.2.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

Router(config-if)#exit
Router(config)#int Se3/0
Router(config-if)#ip address 10.0.0.6 255.255.255.252
Router(config-if)#no shutdown

%LINK-5-CHANGED: Interface Serial3/0, changed state to down
Router(config-if)#exit
Router(config)#router ospf 1
Router(config-router)#network 192.168.2.0 0.0.0.255 Area 2
Router(config-router)#network 10.0.0.4 0.0.0.3 Area 0
Router(config-router)#exit
Router(config)#
%LINK-5-CHANGED: Interface Serial3/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial3/0, changed state to up

01:11:01: %OSPF-5-ADJCHG: Process 1, Nbr 192.168.0.1 on Serial3/0 from LOADING to FULL, Loading Done
```

- **Path Flow:**

  - PC1 (Traffic Signal) sends data to PC0 (Control Center): Data travels from Switch1 → Router1 → Router0 → PC0.
  - PC0 can push commands to PC3 (Emergency Node) through Router0 → Router2 → Switch2 → PC3.

➢ Applications and Use Cases

- **Real-Time Traffic Monitoring:** Control Center receives live data from cameras and sensors.
- **Dynamic Signal Control:** Control Center can change signal timings based on traffic flow.
- **Emergency Routing:** Emergency vehicles are prioritized by triggering green signals.
- **Environmental Monitoring:** Sensors monitor pollution levels and relay data for analysis.

➢ Conclusion:

This Smart Traffic Management System prototype demonstrates how modern networking principles can be applied to real-world smart city applications. By incorporating layered design, IP segmentation, and inter-device communication, this network provides a scalable foundation for intelligent urban traffic control and emergency response systems.